

Isabel Allum

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EDUCATION

Columbia University

M.S. Computer Science, Machine Learning Track - GPA: 3.998

New York, NY

Expected Dec 2026

Courses: Machine Learning, Reinforcement Learning, Artificial Intelligence, Natural Language Processing

University of Chicago

B.A. Economics, Specialization in Data Science - GPA: 3.852

Chicago, IL

2019–2023

Honors: Cum Laude, Dean's List (2020, 2022)

Courses: Econometrics & Machine Learning, Big Data, Data Science Clinic, Data Construction & Interpretation

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, PyTorch, scikit-learn), R (tidyverse, ggplot2), SQL, Stata

Data Tools: PySpark, Databricks, Git/GitHub, Google Cloud Platform

Methods: Causal Inference (DiD, RD, IV), Regression, Forecasting, Clustering, Tree-Based Models, Time Series Analysis

EXPERIENCE

Computer Science Researcher

June 2026 – August 2026

Columbia University

New York, NY

- Designed post-level importance-scoring formulas for an AI memory system, ranking topics from a campus-wide platform with weighted recency, velocity, engagement, and persistence components, replacing the prior topic-level method.
- Built an analysis environment to compare post- and topic-level formulas, uncovering systematic scale differences.

Economic Data Associate

Jul 2023 – Jun 2025

Charles River Associates

Chicago, IL

- Conducted statistical modeling and end-to-end data analysis on terabyte-scale datasets using PySpark (Databricks), R, and Stata to support 10+ multi-million dollar antitrust and litigation cases, while mentoring junior analysts.
- Engineered reproducible data validation and feature engineering workflows to generate analysis-ready datasets from raw client transaction data and public economic data.
- Applied econometric methods including regression modeling, panel analysis, robustness checks, and hypothesis testing to quantify competitive effects and evaluate but-for counterfactual scenarios.
- Communicated technical findings through data visualizations and written reports for broad, non-technical audiences while ensuring analytical defensibility under legal scrutiny.

PROJECTS

Bike Demand Prediction (Personal Project)

Jul 2026

- Built a geospatial data pipeline integrating Citibike trips, Census/LEHD LODES employment data, and NYC Open Data (bike lanes, parks, schools, zoning) using H3 hexagonal indexing across 7 million records.
- Benchmarked linear, negative binomial, and XGBoost models to predict bike traffic between locations, tuning the final model via group-aware grid search to achieve an R^2 of 0.67 on held-out routes.

Dinner Party Planner Web Application

Sep 2025 – Dec 2025

- Designed and implemented a normalized relational database schema in PostgreSQL to efficiently store and query users, events, recipes, and ratings across multiple related tables.
- Built a Flask web application deployed on Google Cloud Compute Engine enabling real-time user interaction using SQL-backed recipe search, rating updates, and event management.

Perpetual (Data Science Consulting)

Jan 2023 – Mar 2023

- Co-developed a weighted k-means spatial optimization model in Python to identify efficient distribution and collection hubs for reusable foodware across two cities; presented recommendations to nonprofit leadership.
- Integrated Google and Yelp API data, performed feature engineering, and visualized geographic demand.